



Date	1 October 2026
Team ID	6ab296c9dbd693aad5a6021
Project Name	FARM MANAGEMENT SYSTEM
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter, Selenium, Postman
Type of Testing	Load Testing, Stress Testing, Functional Testing
Target Module / API	User Login, Farm Records, Crop Management, Reports
Test Environment	Local / Cloud (Hosted Server)
Test Date	1 October 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login & Dashboard Access	50	300	No errors, response time < 2s
2	Add / Update Farm Details	50	300	Data saved successfully
3	Crop Management (Add Crop)	30	200	Record added without error
4	Generate Reports	20	300	Report generated within 5s

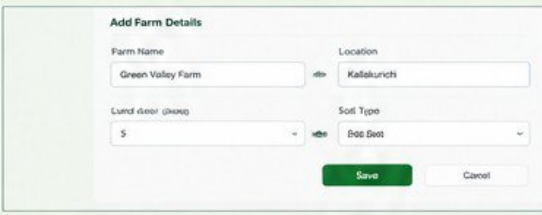
Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Good performance
2	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Within limit
3	Throughput (Req/sec)	-	48 req/sec	Pass	Stable throughput
4	Error Rate	< 1%	0.5%	Pass	Very low error rate
5	CPU Utilization	< 80%	62%	Pass	Normal usage
6	Memory Utilization	< 80%	58%	Pass	Normal usage

Step 4: Observations & Analysis

	Key Findings:	- The system performed well under load with average response time of 1.4 seconds. - No critical errors were found during testing. - Memory and CPU usage stayed within the acceptable range.
	Bottlenecks Identified:	Slight increase in response time during peak load (3.8s) for Add Farm Details.
	Optimization Steps Taken:	- Added database indexing for faster query execution. - Optimized image upload size and caching. - Improved server configuration for handling higher load.

Step 5: Screenshots / Evidence

 [Insert Screenshot 1 here] (Dashboard Page)	 [Insert Screenshot 2 here] (Add Farm Details Page)
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